Experiment No: 1

Aim: To design and develop a Library Management website with a book catalog and search, user and admin login, book issue/return, fine calculation and study material pages using HTML, CSS and JavaScript.

Sample Code:

login.html

```
<!DOCTYPE html>
<html>
<head>
<title>Library Login</title>
<style>
body { font-family: Arial; margin: 30px; }
input, select, button { margin: 5px 0; padding: 5px; }
</style>
</head>
<body>
<h3>Library Login</h3>
<form onsubmit="return login()">
  Role: <select
id="role"><option>Admin</option><option>Student</option></select><br>
  Username: <input type="text" id="user" required><br>
  Password: <input type="password" id="pass" required><br>
  <button type="submit">Login</button>
</form>
<script>
function login() {
  var role = document.getElementById("role").value;
  var u = document.getElementById("user").value;
  var p = document.getElementById("pass").value;
  if ((role == "Admin" && u == "admin" && p == "admin123") ||
      (role == "Student" && u == "student" && p == "stud123")) {
    localStorage.setItem("role", role);
    window.location = "library.html";
  } else {
    alert("Invalid credentials");
  }
  return false;
}
</script>
</body>
</html>
```

library.html

```
<!DOCTYPE html>
<html>
<head>
<title>Library</title>
<style>
body { font-family: Arial; margin: 30px; }
table { border-collapse: collapse; width: 100%; margin-top: 10px; }
th, td { border: 1px solid #555; padding: 6px; text-align: left; }
input, button { padding: 5px; margin: 3px; }
</style>
</head>
<body>
<h3>Library Catalog</h3>
<input type="text" id="search" placeholder="Search by title or author"
onkeyup="show()">
<div id="adminBox">
  <input id="t" placeholder="Title"> <input id="a" placeholder="Author">
  <button onclick="addBook()">Add Book</button>
</div>
```

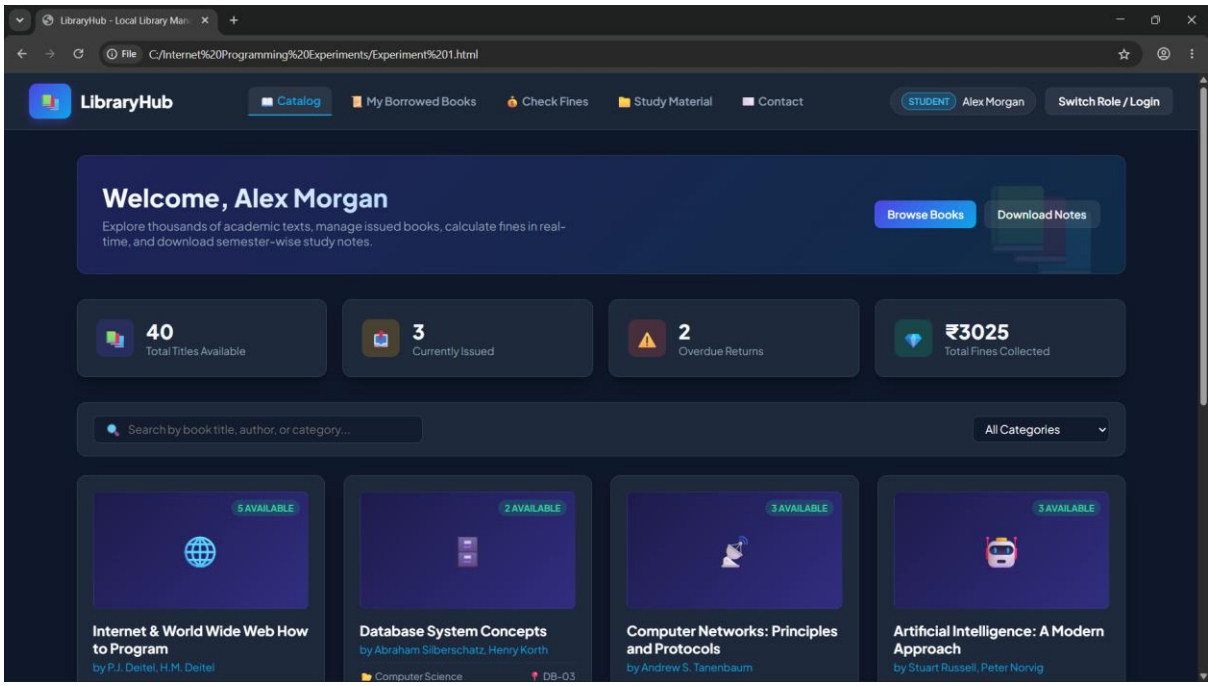
```
<table>
  <thead><tr><th>ID</th><th>Title</th><th>Author</th><th>Status</th>
  <th>Issue Date</th><th>Due Date</th><th>Fine
(Rs)</th><th>Action</th></tr></thead>
  <tbody id="list"></tbody>
</table>
<h3>Study Material</h3>
<select id="yr"><option>1</option><option>2</option><option>3</option></select>
<select id="sem"><option>1</option><option>2</option></select>
<button onclick="material()">View Material</button>
<p id="mat"></p>
<script>
var books = [
  {id:1, title:"Internet Programming", author:"Deitel", issued:false},
  {id:2, title:"Database Systems", author:"Korth", issued:false},
  {id:3, title:"Computer Networks", author:"Tanenbaum", issued:false}
];
var isAdmin = localStorage.getItem("role") == "Admin";
if (!isAdmin) document.getElementById("adminBox").style.display = "none";

function show() {
  var q = document.getElementById("search").value.toLowerCase();
  var rows = "";
  books.forEach(function(b) {
    if ((b.title + b.author).toLowerCase().indexOf(q) < 0) return;
    var fine = b.issued ? calcFine(b) : 0;
    rows += "<tr><td>" + b.id + "</td><td>" + b.title + "</td><td>" + b.author +
      "</td><td>" + (b.issued ? "Issued" : "Available") + "</td><td>" +
(b.issueDate || "-") +
      "</td><td>" + (b.dueDate || "-") + "</td><td>" + fine + "</td><td>" +
      (isAdmin ? (b.issued ? "<button onclick='returnBook(" + b.id +
")'>Return</button>"
                           : "<button onclick='issueBook(" + b.id +
")'>Issue</button>") : "") +
      "</td></tr>";
  });
  document.getElementById("list").innerHTML = rows;
}
function addBook() {
  books.push({id: books.length + 1, title: t.value, author: a.value, issued:
false});
  show();
}
function issueBook(id) {
  var b = books[id - 1], d = new Date(), due = new Date();
  due.setDate(d.getDate() + 14);
  b.issued = true; b.issueDate = d.toDateString(); b.dueDate = due.toDateString();
  b.due = due; show();
}
function calcFine(b) {
  var late = Math.floor((new Date() - b.due) / 86400000);
  return late > 0 ? late * 5 : 0;
}
function returnBook(id) {
  var b = books[id - 1];
  alert("Book returned. Fine = Rs " + calcFine(b));
  b.issued = false; b.issueDate = b.dueDate = null; show();
}
function material() {
  document.getElementById("mat").innerHTML =
    "Year " + yr.value + " Semester " + sem.value + " notes: <a href='#'>Download
PDF</a>";
}
show();
</script>
</body>
</html>
```

Procedure:

1. Open a text editor and create the file login.html with the login form and its validation script.
2. Create the file library.html containing the book catalog table, search box, admin controls and study material section.
3. Write JavaScript functions to add a book, issue a book (14 days due date), return a book and calculate the fine (Rs 5 per late day).
4. Save both files in the same folder and open login.html in a web browser.
5. Log in as Admin (admin / admin123) and test adding, issuing and returning books.
6. Log in as Student (student / stud123), search for a book, check the fine and view the study material.

Output:



Result:

Thus the Library Management website was designed and implemented successfully using HTML, CSS and JavaScript.

Experiment No: 2

Aim: To design and develop an online hotel room booking website with room availability check, booking, modification and cancellation (allowed up to 3 days before arrival) and booking history using HTML, CSS and JavaScript.

Sample Code:

hotel.html

```
<!DOCTYPE html>
<html>
<head>
<title>Hotel Room Booking</title>
<style>
body { font-family: Arial; margin: 30px; }
table { border-collapse: collapse; width: 100%; margin: 10px 0; }
th, td { border: 1px solid #555; padding: 6px; text-align: left; }
input, select, button { padding: 5px; margin: 3px; }
</style>
</head>
<body>
<h3>Hotel Room Booking</h3>
<h4>Room Types</h4>
<table id="rooms"></table>

<h4>Book a Room</h4>
<form onsubmit="return book()">
  Name: <input id="name" required>
  Room Type: <select id="type"></select>
  Arrival: <input type="date" id="arrive" required>
  Departure: <input type="date" id="depart" required>
  Guests: <input type="number" id="guests" min="1" max="4" value="1">
  <button type="submit">Book Now</button>
</form>

<h4>My Booking History</h4>
<table id="history"></table>

<script>
var rooms = [
  {type:"Single", price:1500, qty:5},
  {type:"Double", price:2500, qty:4},
  {type:"Deluxe", price:4000, qty:3},
  {type:"Suite",  price:6500, qty:2}
];
var bookings = [];

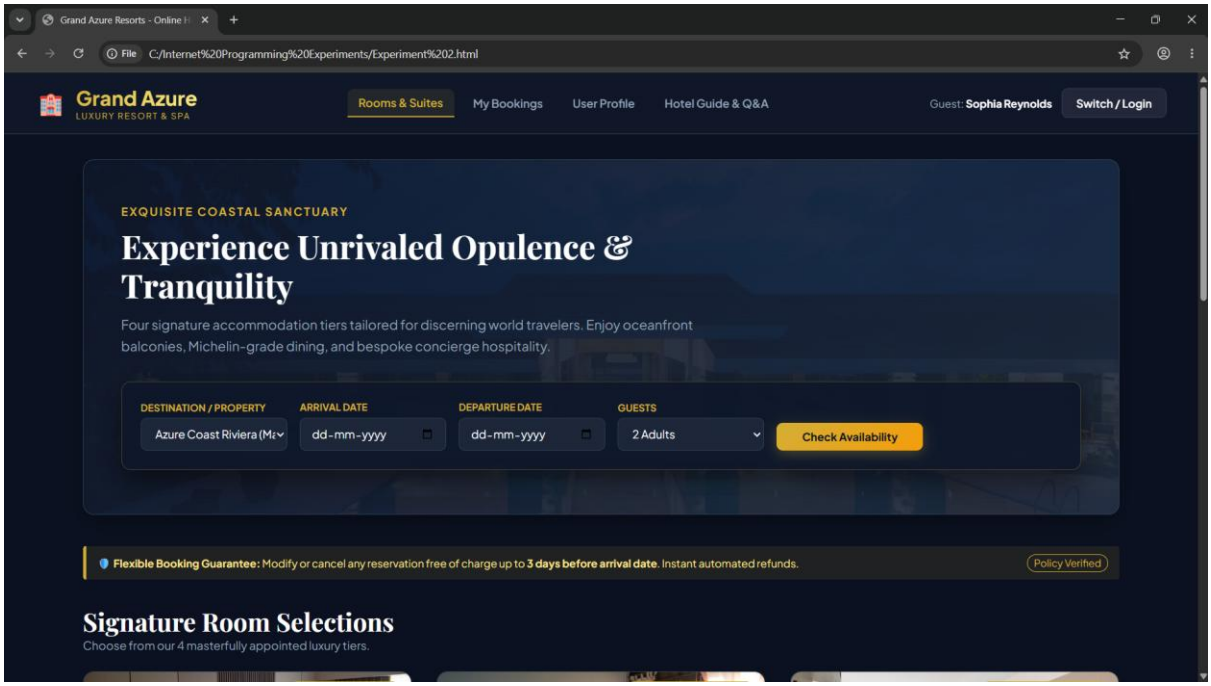
function showRooms() {
  var h = "<tr><th>Type</th><th>Price/Night</th><th>Available</th></tr>";
  var opt = "";
  rooms.forEach(function(r) {
    h += "<tr><td>" + r.type + "</td><td>Rs " + r.price + "</td><td>" + r.qty +
"</td></tr>";
    opt += "<option>" + r.type + "</option>";
  });
  document.getElementById("rooms").innerHTML = h;
  document.getElementById("type").innerHTML = opt;
}
function findRoom(t) {
  return rooms.filter(function(r) { return r.type == t; })[0];
}
function book() {
  var room = findRoom(type.value);
  var a = new Date(arrive.value), d = new Date(depart.value);
  if (d <= a) { alert("Departure must be after arrival"); return false; }
```

```
  if (room.qty == 0) { alert("No rooms available"); return false; }
  var nights = (d - a) / 86400000;
  room.qty--;
  bookings.push({id: bookings.length + 1, name: name.value, type: type.value,
    arrive: arrive.value, depart: depart.value, total: nights * room.price, status:
"Booked"});
  showRooms(); showHistory();
  alert("Booking confirmed. Total = Rs " + nights * room.price);
  return false;
}
function daysLeft(b) {
  return (new Date(b.arrive) - new Date()) / 86400000;
}
function cancelBooking(i) {
  var b = bookings[i];
  if (daysLeft(b) < 3) { alert("Cancellation allowed only 3 days before arrival");
return; }
  b.status = "Cancelled"; findRoom(b.type).qty++;
  showRooms(); showHistory();
}
function changeDate(i) {
  var b = bookings[i];
  if (daysLeft(b) < 3) { alert("Changes allowed only 3 days before arrival");
return; }
  var n = prompt("Enter new arrival date (YYYY-MM-DD)", b.arrive);
  if (n) { b.arrive = n; showHistory(); }
}
function showHistory() {
  var h =
"<tr><th>ID</th><th>Name</th><th>Room</th><th>Arrival</th><th>Departure</th>" +
          "<th>Total</th><th>Status</th><th>Action</th></tr>";
  bookings.forEach(function(b, i) {
    h += "<tr><td>" + b.id + "</td><td>" + b.name + "</td><td>" + b.type +
"</td><td>" +
      b.arrive + "</td><td>" + b.depart + "</td><td>" + b.total + "</td><td>" +
b.status + "</td><td>" +
      (b.status == "Booked" ? "<button onclick='changeDate(" + i +
")'>Change</button>" +
      "<button onclick='cancelBooking(" + i + ")'>Cancel</button>" : "") +
"</td></tr>";
  });
  document.getElementById("history").innerHTML = h;
}
showRooms(); showHistory();
</script>
</body>
</html>
```

Procedure:

1. Create the file hotel.html and design the page layout with CSS.
2. Define the four room types (Single, Double, Deluxe, Suite) with price and quantity, and display them in a table.
3. Create the booking form and write the book() function to check availability, calculate the total amount and store the booking.
4. Write the cancelBooking() and changeDate() functions with the rule that changes are allowed only 3 days before the arrival date.
5. Display the booking history table for the user.
6. Open hotel.html in a browser and test booking, availability update, modification and cancellation.

Output:



Result:

Thus the online hotel room booking website was designed and implemented successfully using HTML, CSS and JavaScript.

Experiment No: 3

Aim: To design and develop an online bus ticket booking website with bus search by route and date, seat selection, passenger details, booking confirmation and ticket cancellation using HTML, CSS and JavaScript.

Sample Code:

bus.html

```
<!DOCTYPE html>
<html>
<head>
<title>Bus Ticket Booking</title>
<style>
body { font-family: Arial; margin: 30px; }
table { border-collapse: collapse; width: 100%; margin: 10px 0; }
th, td { border: 1px solid #555; padding: 6px; text-align: left; }
.seat { width: 40px; height: 35px; margin: 3px; background: #c8e6c9; border: 1px
solid #333; cursor: pointer; }
.booked { background: #ef9a9a; cursor: not-allowed; }
.selected { background: #fff176; }
input, select, button { padding: 5px; margin: 3px; }
</style>
</head>
<body>
<h3>Bus Ticket Booking</h3>
<h4>Search Bus</h4>
From: <select
id="from"><option>Chennai</option><option>Madurai</option><option>Coimbatore</optio
n></select>
To: <select
id="to"><option>Madurai</option><option>Chennai</option><option>Coimbatore</option>
</select>
Date: <input type="date" id="date">
<button onclick="searchBus()">Search</button>
<table id="busList"></table>

<div id="seatBox" style="display:none">
  <h4>Select Seats</h4>
  <div id="seats"></div>
  <h4>Passenger Details</h4>
  Name: <input id="pname"> Age: <input type="number" id="page" style="width:60px">
  Phone: <input id="pphone">
  <button onclick="confirmBooking()">Pay and Book</button>
</div>

<h4>My Bookings</h4>
<table id="tickets"></table>

<script>
var buses = [
  {id:1, name:"Volvo AC", from:"Chennai", to:"Madurai", time:"21:00", fare:700,
booked:[2,5,9]},
  {id:2, name:"Sleeper Coach", from:"Chennai", to:"Madurai", time:"22:30",
fare:900, booked:[1,3]},
  {id:3, name:"Express", from:"Madurai", to:"Chennai", time:"20:00", fare:450,
booked:[]}
];
var tickets = [], current = null, chosen = [];

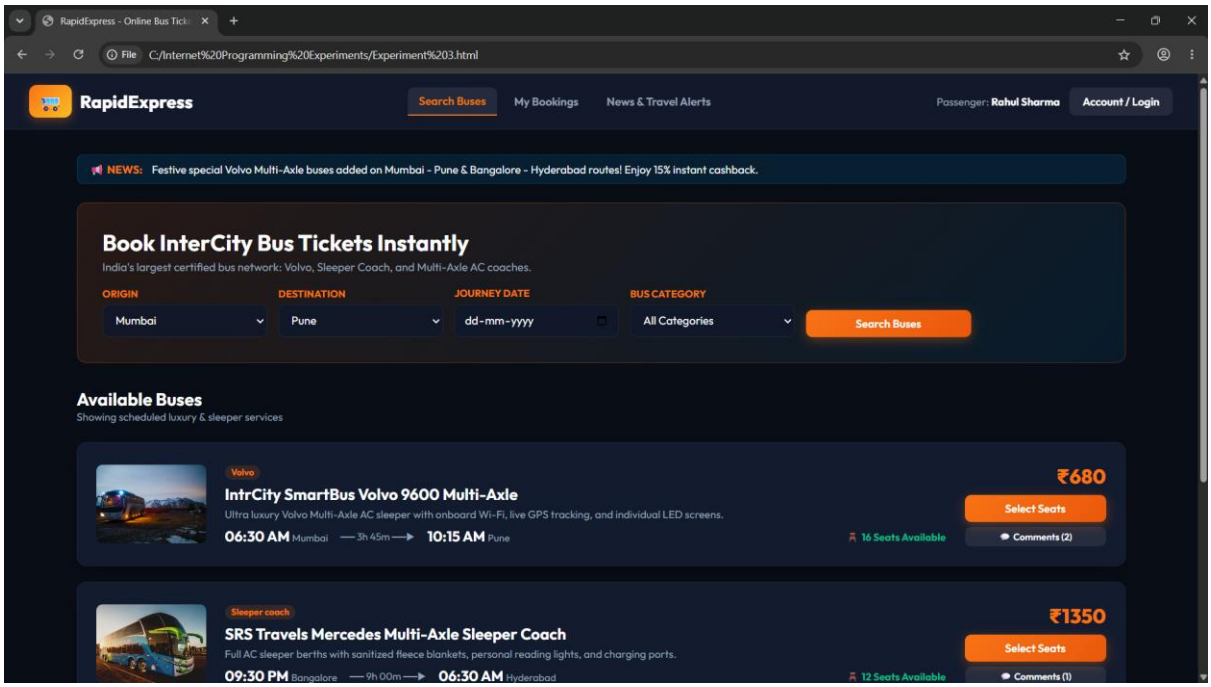
function searchBus() {
  var h = "<tr><th>Bus</th><th>Time</th><th>Fare</th><th>Seats
Left</th><th></th></tr>";
  buses.forEach(function(b) {
    if (b.from == from.value && b.to == to.value)
```

```
      h += "<tr><td>" + b.name + "</td><td>" + b.time + "</td><td>Rs " + b.fare +
"</td><td>" +
        (20 - b.booked.length) + "</td><td><button onclick='pickBus(" + b.id +
")'>Select</button></td></tr>";
  });
  document.getElementById("busList").innerHTML = h;
}
function pickBus(id) {
  current = buses[id - 1]; chosen = [];
  var h = "";
  for (var i = 1; i <= 20; i++) {
    var isB = current.booked.indexOf(i) >= 0;
    h += "<button class='seat " + (isB ? "booked" : "") + "' id='s" + i + "' " +
         (isB ? "disabled" : "onclick='toggle(" + i + ")'") + ">" + i +
"</button>";
    if (i % 4 == 0) h += "<br>";
  }
  document.getElementById("seats").innerHTML = h;
  document.getElementById("seatBox").style.display = "block";
}
function toggle(n) {
  var i = chosen.indexOf(n);
  if (i >= 0) chosen.splice(i, 1); else chosen.push(n);
  document.getElementById("s" + n).classList.toggle("selected");
}
function confirmBooking() {
  if (chosen.length == 0 || !pname.value) { alert("Select seats and enter
details"); return; }
  tickets.push({id: tickets.length + 1, name: pname.value, bus: current.name, date:
date.value,
    seats: chosen.join(","), amount: chosen.length * current.fare, status:
"Confirmed"});
  chosen.forEach(function(s) { current.booked.push(s); });
  alert("Booking confirmed. A confirmation email has been sent.");
  document.getElementById("seatBox").style.display = "none";
  showTickets(); searchBus();
}
function cancelTicket(i) {
  var t = tickets[i];
  t.status = "Cancelled";
  buses.forEach(function(b) {
    if (b.name == t.bus) t.seats.split(",").forEach(function(s) {
      b.booked.splice(b.booked.indexOf(Number(s)), 1);
    });
  });
  showTickets(); searchBus();
}
function showTickets() {
  var h =
"<tr><th>ID</th><th>Passenger</th><th>Bus</th><th>Date</th><th>Seats</th><th>Amount
</th><th>Status</th><th></th></tr>";
  tickets.forEach(function(t, i) {
    h += "<tr><td>" + t.id + "</td><td>" + t.name + "</td><td>" + t.bus +
"</td><td>" + t.date +
      "</td><td>" + t.seats + "</td><td>" + t.amount + "</td><td>" + t.status +
"</td><td>" +
      (t.status == "Confirmed" ? "<button onclick='cancelTicket(" + i +
")'>Cancel</button>" : "") + "</td></tr>";
  });
  document.getElementById("tickets").innerHTML = h;
}
</script>
</body>
</html>
```

Procedure:

1. Create the file bus.html and add the search form with source, destination and date fields.
2. Store the bus details (name, route, time, fare and already booked seats) in a JavaScript array.
3. Write searchBus() to list the buses matching the selected route.
4. Generate the seat layout dynamically; booked seats are disabled and selected seats are highlighted.
5. Create the passenger details form and write confirmBooking() to save the ticket and update seat availability.
6. Write cancelTicket() to cancel a booking and release the seats, then test the page in a browser.

Output:



Result:

Thus the online bus ticket booking website was designed and implemented successfully using HTML, CSS and JavaScript.

Experiment No: 4

Aim: To design and develop an online train reservation system with login, train search, ticket booking with PNR generation, PNR status check and ticket cancellation with refund calculation using HTML, CSS and JavaScript.

Sample Code:

train.html

```
<!DOCTYPE html>
<html>
<head>
<title>Train Reservation</title>
<style>
body { font-family: Arial; margin: 30px; }
table { border-collapse: collapse; width: 100%; margin: 10px 0; }
th, td { border: 1px solid #555; padding: 6px; text-align: left; }
input, select, button { padding: 5px; margin: 3px; }
.box { border: 1px solid #999; padding: 10px; margin-bottom: 15px; }
</style>
</head>
<body>
<h3>Railway Reservation System</h3>

<div class="box" id="loginBox">
  <h4>Login</h4>
  User: <input id="u"> Password: <input type="password" id="p">
  <button onclick="login()">Login</button>
</div>

<div id="main" style="display:none">
  <div class="box">
    <h4>Search Trains</h4>
    From: <input id="src" value="Chennai"> To: <input id="dst" value="Delhi">
    Date: <input type="date" id="jdate">
    <button onclick="search()">Search</button>
    <table id="trainList"></table>
  </div>

  <div class="box" id="bookBox" style="display:none">
    <h4>Passenger Details</h4>
    Name: <input id="pn"> Age: <input type="number" id="pa" style="width:60px">
    Class: <select
id="cls"><option>SL</option><option>3A</option><option>2A</option></select>
    <button onclick="bookTicket()">Pay and Book</button>
  </div>

  <div class="box">
    <h4>PNR Status / Cancel Ticket</h4>
    PNR: <input id="pnr">
    <button onclick="status()">Check Status</button>
    <button onclick="cancel()">Cancel Ticket</button>
    <p id="msg"></p>
  </div>
</div>

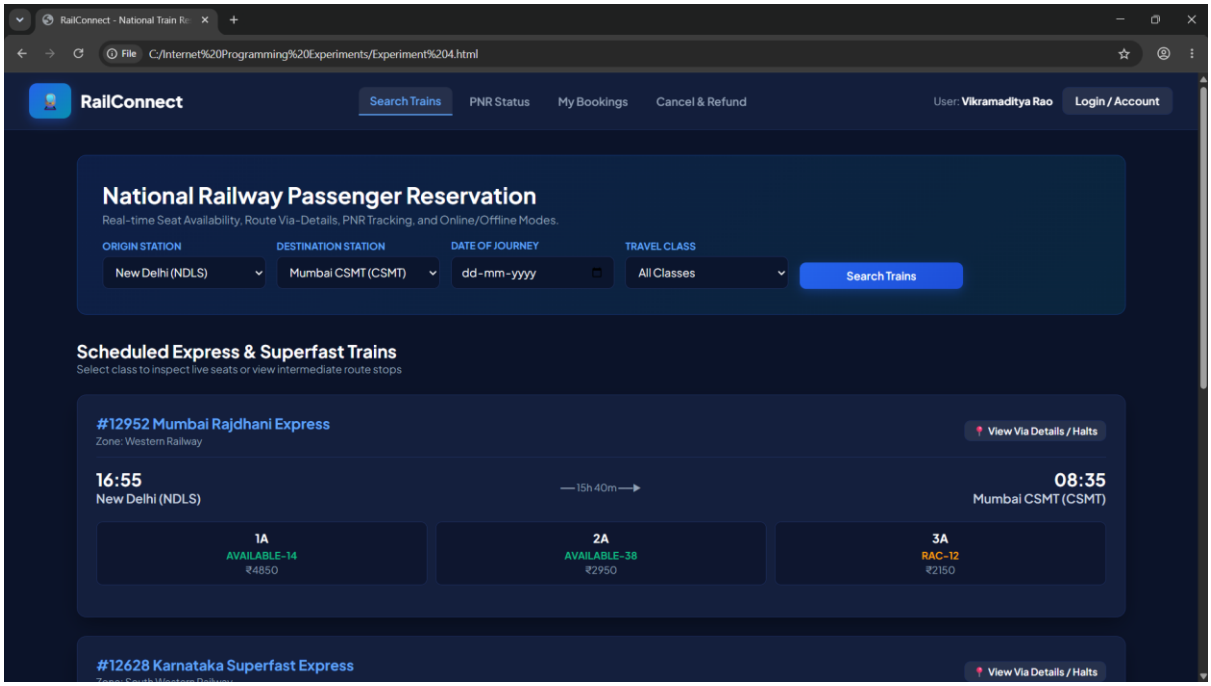
<script>
var trains = [
  {no:12621, name:"Tamil Nadu Express", from:"Chennai", to:"Delhi", fare:{SL:800,
"3A":2100, "2A":3000}, seats:3},
  {no:12615, name:"Grand Trunk Express", from:"Chennai", to:"Delhi", fare:{SL:750,
"3A":1950, "2A":2800}, seats:0}
];
var tickets = {}, chosen = null;
```

```
function login() {
  if (u.value == "user" && p.value == "1234") {
    loginBox.style.display = "none"; main.style.display = "block";
  } else alert("Invalid login");
}
function search() {
  var h =
"<tr><th>No</th><th>Train</th><th>SL</th><th>3A</th><th>2A</th><th>Seats</th><th></
th></tr>";
  trains.forEach(function(t, i) {
    if (t.from == src.value && t.to == dst.value)
      h += "<tr><td>" + t.no + "</td><td>" + t.name + "</td><td>" + t.fare.SL +
"</td><td>" +
        t.fare["3A"] + "</td><td>" + t.fare["2A"] + "</td><td>" + (t.seats > 0 ?
t.seats : "WL") +
        "</td><td><button onclick='pick(" + i + ")'>Book</button></td></tr>";
  });
  trainList.innerHTML = h;
}
function pick(i) { chosen = trains[i]; bookBox.style.display = "block"; }
function bookTicket() {
  var pnrNo = Math.floor(1000000000 + Math.random() * 9000000000);
  var st = chosen.seats > 0 ? "Confirmed" : "Waiting List";
  if (chosen.seats > 0) chosen.seats--;
  tickets[pnrNo] = {name: pn.value, train: chosen.name, cls: cls.value,
                    fare: chosen.fare[cls.value], date: jdate.value, status: st};
  msg.innerHTML = "Ticket booked. PNR: " + pnrNo + " Status: " + st;
}
function status() {
  var t = tickets[pnr.value];
  msg.innerHTML = t ? t.name + " | " + t.train + " | " + t.cls + " | " + t.date + "
| " + t.status
                    : "Invalid PNR";
}
function cancel() {
  var t = tickets[pnr.value];
  if (!t || t.status == "Cancelled") { msg.innerHTML = "Invalid or already
cancelled PNR"; return; }
  var days = (new Date(t.date) - new Date()) / 86400000;
  var percent = days > 2 ? 0.75 : (days > 0.5 ? 0.5 : 0.25);
  t.status = "Cancelled";
  msg.innerHTML = "Ticket cancelled. Refund = Rs " + (t.fare * percent);
}
</script>
</body>
</html>
```

Procedure:

1. Create the file train.html with the login section, search section, booking section and PNR section.
2. Store train details (number, name, route, class-wise fare and seats) in a JavaScript array.
3. Write login() to validate the user and then display the reservation page.
4. Write search() to list trains for the entered route and bookTicket() to generate a random PNR number and set the status as Confirmed or Waiting List.
5. Write status() to display ticket details for a PNR and cancel() to cancel the ticket and calculate the refund based on the number of days left before the journey.
6. Open train.html in a browser, log in (user / 1234) and test booking, PNR status and cancellation.

Output:



Result:

Thus the online train reservation system was designed and implemented successfully using HTML, CSS and JavaScript.

Experiment No: 5

Aim: To design and develop an online flight reservation system with user login, flight search, ticket booking, my ticket view, ticket cancellation and ticket printing using HTML, CSS and JavaScript.

Sample Code:

flight.html

```
<!DOCTYPE html>
<html>
<head>
<title>Flight Reservation</title>
<style>
body { font-family: Arial; margin: 30px; }
table { border-collapse: collapse; width: 100%; margin: 10px 0; }
th, td { border: 1px solid #555; padding: 6px; text-align: left; }
input, select, button { padding: 5px; margin: 3px; }
.box { border: 1px solid #999; padding: 10px; margin-bottom: 15px; }
</style>
</head>
<body>
<h3>Flight Reservation System</h3>

<div class="box" id="loginBox">
  <h4>Login</h4>
  Email: <input id="email"> Password: <input type="password" id="pwd">
  <button onclick="login()">Login</button>
  <a href="#" onclick="alert('Reset link sent to your email')">Forgot Password?</a>
</div>

<div id="main" style="display:none">
  <div class="box">
    <h4>Search Flight</h4>
    From: <input id="from" value="Chennai"> To: <input id="to" value="Mumbai">
    Date: <input type="date" id="fdate">
    Class: <select
id="fclass"><option>Economy</option><option>Business</option></select>
    <button onclick="search()">Search</button>
    <table id="flightList"></table>
  </div>

  <div class="box" id="bookBox" style="display:none">
    <h4>Passenger Details and Payment</h4>
    Name: <input id="pname"> Age: <input type="number" id="page"
style="width:60px">
    Seat No: <input id="seat" style="width:60px">
    Card No: <input id="card" maxlength="16">
    <button onclick="book()">Pay and Book</button>
  </div>

  <div class="box">
    <h4>My Tickets</h4>
    <table id="myTickets"></table>
  </div>
</div>

<script>
var flights = [
  {no:"AI-101", from:"Chennai", to:"Mumbai", dep:"06:30", arr:"08:35",
price:{Economy:4500, Business:11000}},
  {no:"6E-202", from:"Chennai", to:"Mumbai", dep:"13:15", arr:"15:20",
price:{Economy:3900, Business:9800}},
  {no:"SG-303", from:"Chennai", to:"Delhi",  dep:"09:00", arr:"11:45",
price:{Economy:5200, Business:12500}}
```

```
];
var tickets = [], chosen = null;

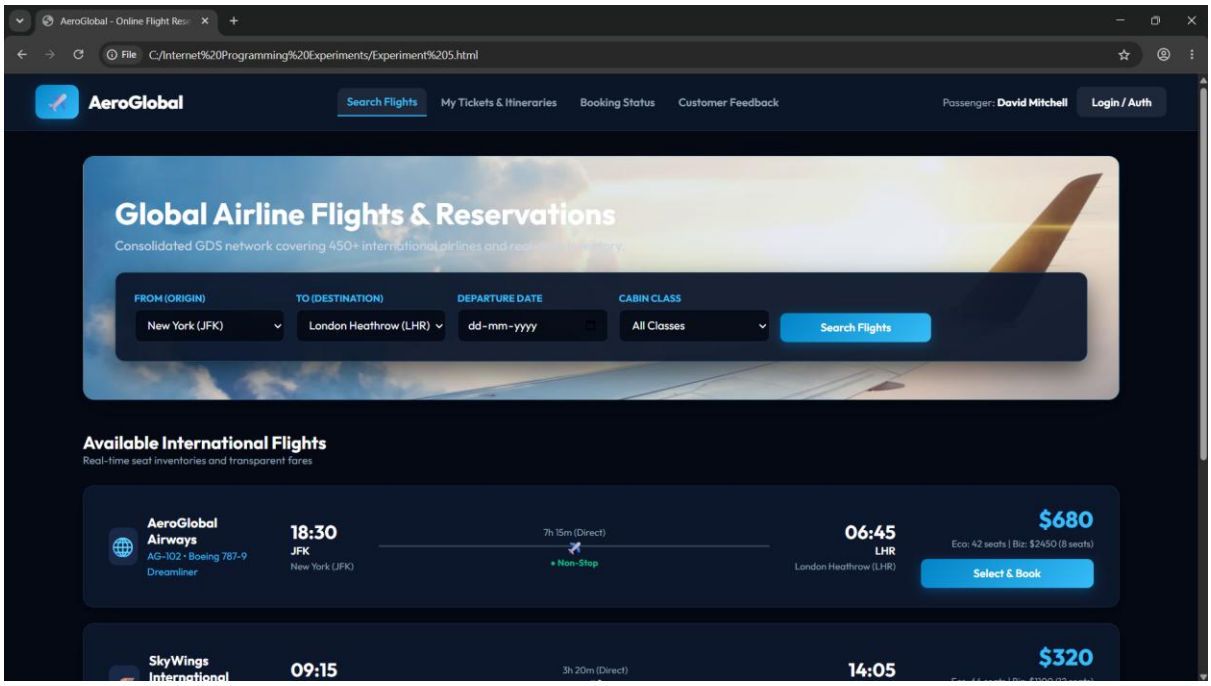
function login() {
  if (email.value == "user@mail.com" && pwd.value == "1234") {
    loginBox.style.display = "none"; main.style.display = "block";
  } else alert("Invalid email or password");
}
function search() {
  var h =
"<tr><th>Flight</th><th>Departure</th><th>Arrival</th><th>Fare</th><th></th></tr>";
  flights.forEach(function(f, i) {
    if (f.from == from.value && f.to == to.value)
      h += "<tr><td>" + f.no + "</td><td>" + f.dep + "</td><td>" + f.arr +
"</td><td>Rs " +
        f.price[fclass.value] + "</td><td><button onclick='pick(" + i +
")'>Book</button></td></tr>";
  });
  flightList.innerHTML = h;
}
function pick(i) { chosen = flights[i]; bookBox.style.display = "block"; }
function book() {
  if (!pname.value || card.value.length != 16) { alert("Enter valid details");
return; }
  tickets.push({id: "TKT" + (1000 + tickets.length), name: pname.value, flight:
chosen.no,
    route: chosen.from + " - " + chosen.to, date: fdate.value, cls: fclass.value,
    seat: seat.value, fare: chosen.price[fclass.value], status: "Booked"});
  alert("Payment successful. E-ticket sent to your email.");
  bookBox.style.display = "none"; showTickets();
}
function cancelTicket(i) { tickets[i].status = "Cancelled"; showTickets(); }
function printTicket(i) {
  var t = tickets[i];
  var w = window.open("", "", "width=500,height=400");
  w.document.write("<h3>E-Ticket</h3>Ticket: " + t.id + "<br>Passenger: " + t.name
+
    "<br>Flight: " + t.flight + " (" + t.route + ")<br>Date: " + t.date +
    "<br>Class: " + t.cls + "<br>Seat: " + t.seat + "<br>Fare: Rs " + t.fare);
  w.print();
}
function showTickets() {
  var h =
"<tr><th>Ticket</th><th>Passenger</th><th>Flight</th><th>Date</th><th>Fare</th><th>
Status</th><th>Action</th></tr>";
  tickets.forEach(function(t, i) {
    h += "<tr><td>" + t.id + "</td><td>" + t.name + "</td><td>" + t.flight +
"</td><td>" + t.date +
      "</td><td>" + t.fare + "</td><td>" + t.status + "</td><td>" +
      (t.status == "Booked" ? "<button onclick='printTicket(" + i +
")'>Print</button>" +
      "<button onclick='cancelTicket(" + i + ")'>Cancel</button>" : "") +
"</td></tr>";
  });
  myTickets.innerHTML = h;
}
</script>
</body>
</html>
```

Procedure:

1. Create the file flight.html with the login page including a Forgot Password link.
2. Store flight details (flight number, route, timings and class-wise price) in a JavaScript array.

3. Write search() to display the flights matching the source, destination and class.
4. Create the passenger details and payment form, and write book() to validate the card number and generate the e-ticket.
5. Write showTickets(), cancelTicket() and printTicket() to manage the My Ticket page.
6. Open flight.html in a browser, log in (user@mail.com / 1234) and test searching, booking, cancelling and printing a ticket.

Output:



Result:

Thus the online flight reservation system was designed and implemented successfully using HTML, CSS and JavaScript.

Experiment No: 6

Aim: To design and develop an online cake ordering website with cake search, category and price filters, cart, shipping address, order placement and admin product and order management using HTML, CSS and JavaScript.

Sample Code:

cake.html

```
<!DOCTYPE html>
<html>
<head>
<title>Online Cake Shop</title>
<style>
body { font-family: Arial; margin: 30px; }
table { border-collapse: collapse; width: 100%; margin: 10px 0; }
th, td { border: 1px solid #555; padding: 6px; text-align: left; }
input, select, button { padding: 5px; margin: 3px; }
.box { border: 1px solid #999; padding: 10px; margin-bottom: 15px; }
</style>
</head>
<body>
<h3>Online Cake Shop</h3>
Login as:
<select id="role" onchange="setRole()">
  <option>Customer</option><option>Admin</option>
</select>

<div class="box">
  <h4>Search and Filter</h4>
  <input id="q" placeholder="Search cake" onkeyup="show()">
  Category:
  <select id="cat" onchange="show()">

<option>All</option><option>Birthday</option><option>Wedding</option><option>Annive
rsary</option>
  </select>
  Max Price: <input type="number" id="maxp" value="2000" onchange="show()"
style="width:80px">
  <table id="cakeList"></table>
</div>

<div class="box" id="adminBox" style="display:none">
  <h4>Admin - Add Cake</h4>
  Name: <input id="cn"> Category: <input id="cc"> Price: <input type="number"
id="cp">
  Description: <input id="cd">
  <button onclick="addCake()">Post Cake</button>
</div>

<div class="box" id="custBox">
  <h4>My Cart</h4>
  <table id="cartList"></table>
  <p id="total"></p>
  Shipping Address: <input id="addr" size="40">
  <button onclick="placeOrder()">Place Order</button>
</div>

<div class="box">
  <h4>Orders</h4>
  <table id="orderList"></table>
</div>

<script>
var cakes = [
```

```
  {id:1, name:"Chocolate Truffle", cat:"Birthday", price:650, desc:"Rich dark
chocolate cake"},
  {id:2, name:"Black Forest", cat:"Birthday", price:550, desc:"Cherry and cream
layers"},
  {id:3, name:"Three Tier Fondant", cat:"Wedding", price:1800, desc:"Designer
wedding cake"},
  {id:4, name:"Red Velvet", cat:"Anniversary", price:750, desc:"Cream cheese
frosting"}
];
var cart = [], orders = [];

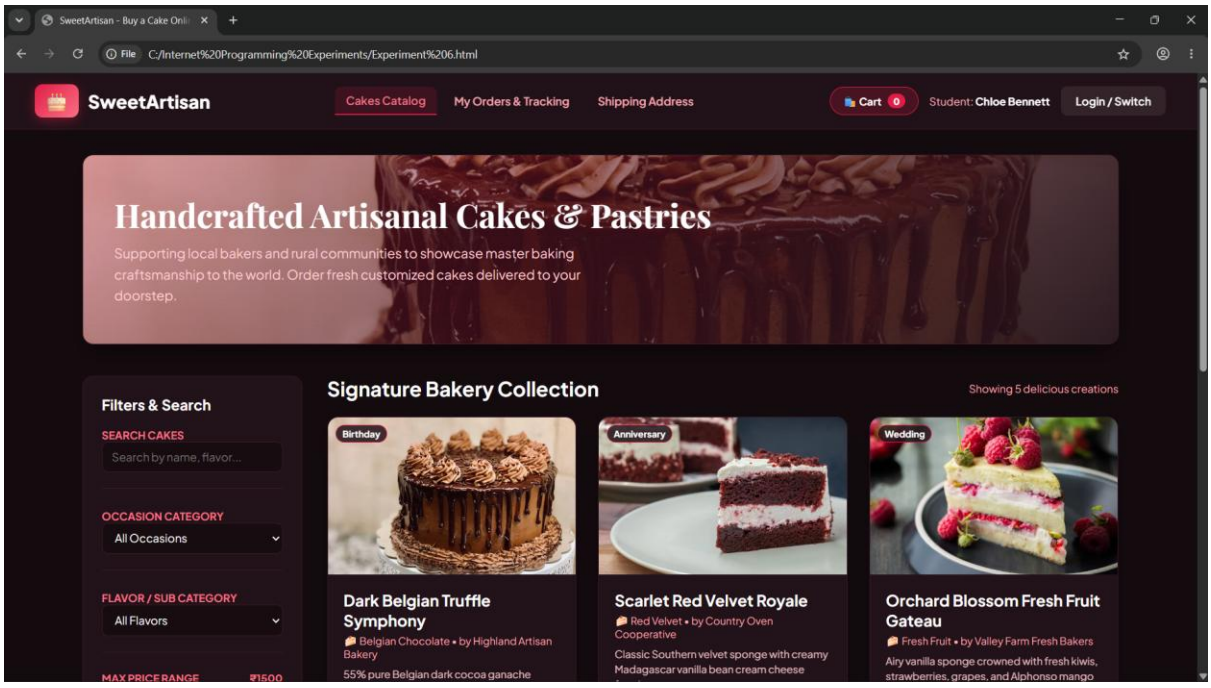
function show() {
  var h =
"<tr><th>Name</th><th>Category</th><th>Description</th><th>Price</th><th></th></tr>
";
  cakes.forEach(function(c) {
    var okName = c.name.toLowerCase().indexOf(q.value.toLowerCase()) >= 0;
    var okCat = cat.value == "All" || cat.value == c.cat;
    if (okName && okCat && c.price <= Number(maxp.value))
      h += "<tr><td>" + c.name + "</td><td>" + c.cat + "</td><td>" + c.desc +
"</td><td>Rs " +
       c.price + "</td><td><button onclick='addCart(" + c.id + ")'>Add to
Cart</button></td></tr>";
  });
  cakeList.innerHTML = h;
}
function addCart(id) { cart.push(cakes[id - 1]); showCart(); }
function removeCart(i) { cart.splice(i, 1); showCart(); }
function showCart() {
  var h = "<tr><th>Cake</th><th>Price</th><th></th></tr>", sum = 0;
  cart.forEach(function(c, i) {
    sum += c.price;
    h += "<tr><td>" + c.name + "</td><td>" + c.price +
         "</td><td><button onclick='removeCart(" + i +
")'>Remove</button></td></tr>";
  });
  cartList.innerHTML = h;
  total.innerHTML = "Total: Rs " + sum;
}
function placeOrder() {
  if (cart.length == 0 || !addr.value) { alert("Add cakes and shipping address");
return; }
  var sum = 0, names = [];
  cart.forEach(function(c) { sum += c.price; names.push(c.name); });
  orders.push({id: orders.length + 1, items: names.join(", "), amount: sum,
               address: addr.value, status: "Order Placed"});
  cart = []; showCart(); showOrders();
}
function updateStatus(i) {
  orders[i].status = document.getElementById("st" + i).value; showOrders();
}
function showOrders() {
  var isAdmin = role.value == "Admin";
  var h =
"<tr><th>ID</th><th>Items</th><th>Amount</th><th>Address</th><th>Status</th></tr>";
  orders.forEach(function(o, i) {
    var st = isAdmin
      ? "<select id='st" + i + "' onchange='updateStatus(" + i + ")'><option>Order
Placed</option>" +
        "<option>Packed</option><option>Out for
Delivery</option><option>Delivered</option></select>"
      : o.status;
    h += "<tr><td>" + o.id + "</td><td>" + o.items + "</td><td>" + o.amount +
"</td><td>" +
         o.address + "</td><td>" + st + "</td></tr>";
  });
  orderList.innerHTML = h;
}
```

```
function addCake() {
  cakes.push({id: cakes.length + 1, name: cn.value, cat: cc.value, price:
Number(cp.value), desc: cd.value});
  show();
}
function setRole() {
  var isAdmin = role.value == "Admin";
  adminBox.style.display = isAdmin ? "block" : "none";
  custBox.style.display = isAdmin ? "none" : "block";
  showOrders();
}
show(); showCart();
</script>
</body>
</html>
```

Procedure:

1. Create the file cake.html and design the page with CSS.
2. Store the cake details (name, category, description and price) in a JavaScript array.
3. Write show() to display the cake list with search, category and price range filters.
4. Write the cart functions (add, remove and total calculation) and placeOrder() to save the order with the shipping address.
5. Write the admin functions to post a new cake and to update the order status.
6. Open cake.html in a browser and test the customer view and the admin view.

Output:



Result:

Thus the online cake ordering website was designed and implemented successfully using HTML, CSS and JavaScript.